# With 750 labelled tweets, does fine-tuning beat a calibrated open LLM?

RESEARCH QUESTION · UNCHANGED

Take **750 labelled OffensEval-TR tweets**. Fine-tune **BERTurk**, or prompt **Gemma-2-9B-it** or **Qwen3-8B** with 8 examples from those 750 and use the other 742 to set a decision threshold. Compare **macro-F1** on the frozen 3,528-tweet test set.

WE CALL IT A NO WHEN

BERTurk's mean over 10 runs is less than **0.02** above the LLM's mean over 3 runs, or a paired bootstrap says the gap could be chance.

| 0.763 | 0.767 | 0.790 | 60 |
|---|---|---|---|
| TF-IDF baseline | BERTurk-750 | Gemma-4-12B | baseline errors |
| frozen test | dev, 10 runs | dev, calibrated | audited |

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# 30,000 train / 1,756 dev / 3,528 frozen test; 19.3% offensive

| FILE / SPLIT | TWEETS | NOT | OFF | OFF % |
|---|---|---|---|---|
| Official training file | 31,756 | 25,625 | 6,131 | 19.3 |
| Our train (seed 42) | 30,000 | 24,208 | 5,792 | 19.3 |
| Our dev | 1,756 | 1,417 | 339 | 19.3 |
| Official test (frozen) | 3,528 | 2,812 | 716 | 20.3 |

OffensEval-TR 2020 (Çöltekin, LREC 2020). Annotations CC BY 2.0. @USER/URL already masked; no extra cleaning. 9 duplicate train texts; 1 train/test overlap.

REAL EXAMPLE · OFF · TRAIN ID 33518

*“Kırmızı ağır karar Allah belanı versin”*

“A red card is a harsh call, God damn you.” Swearing at a decision, not a person; still OFF.

CONTRAST · NOT

*“...annem alacağı arabadan kaç kere vazgeçti ya kahretsin zalım trafik”*

“How many times my mum gave up on buying a car, damn this cruel traffic.” Cursing at traffic is not offensive; target matters.

WHY THESE COUNTS MATTER

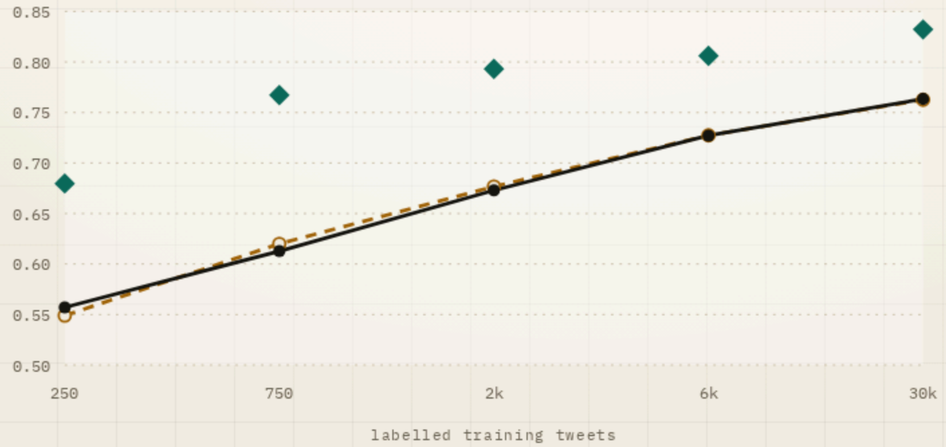
Only 19.3% of training tweets are offensive. A model that only answers “not offensive” looks right at first glance: 80.7% accuracy. Its macro-F1 is 0.447, though, so accuracy alone hides the failure. The test set stays frozen; dev is only for code checks and mid-project numbers, and every system uses the saved subsets (seed 42 + k).

# TF-IDF + LR reaches 0.763 on test; 0.620 with 750 labels

| SYSTEM | TEST MACRO-F1 | OFF P | OFF R | SPREAD |
|---|---|---|---|---|
| Always NOT | 0.444 | n/a | 0.000 | deterministic |
| TF-IDF + LR, 750 | 0.620 ± 0.012 | n/a | n/a | sd over 10 samples |
| TF-IDF + LR, 30,000 | 0.763 | 0.589 | 0.673 | 95% CI [0.745, 0.779] |

The frozen test set was scored once for this deterministic baseline. All method results stay on dev until the final freeze.

**0.620** test at 750 labels
0.613 on dev




Solid: TF-IDF dev. Dashed: TF-IDF test. Diamonds: fine-tuned BERTurk dev.

WHERE THE FLOOR SITS

With 750 labels, the simple baseline scores 0.620 on test and 0.613 on dev. That is the minimum BERTurk and the LLMs must beat. At 30,000 labels, TF-IDF climbs to 0.763; BERTurk scores 0.767 with only 750 labels and 0.832 with 30,000 on dev.

# 275 dev errors; a 60-error audit shows where a word model fails

| FALSE ALARMS (166): CATEGORY | N |
|---|---|
| No insult; spurious word/char cue | 9 |
| Political or religious criticism | 10 |
| Sports criticism or banter | 5 |
| Personal criticism not judged offensive | 3 |
| Sarcasm or irony | 1 |
| Label noise or genuinely ambiguous | 2 |

| MISSED OFFENSIVE (109): CATEGORY | N |
|---|---|
| Implicit insult, no profanity | 8 |
| Masked or abbreviated profanity | 5 |
| Veiled threat or violence | 3 |
| Political or religious insult | 4 |
| Institutional or sports criticism | 3 |
| Label noise or not offensive | 7 |

Counts are in the 60-error audit (30 per direction, seed 42); each table sums to 30.

FALSE ALARM · F3 · 13105

*"Aziz Yıldırım Fenerbahçe'yi padişahlık zannediyor herhalde..."*

Club criticism without a slur.

MISS · M1 · 43325

*"...bundan daha parazit canlı türü yok"*

Implicit insult with no swear word.

MISS · M2 · 15069

*"...bakanlık seçimi kaldı amkk"*

Masked profanity, unseen by the word/char model.

WHAT THE AUDIT SUGGESTS

False alarms often involve conflict and club words and character combinations. Implicit insults and masked profanity are easily missed. About one in seven audited errors is label noise, so the ceiling is below 1.0.

# Both arms run end-to-end; first dev numbers

CONTROLLED CHANGE

The baseline trains its weights on the 750 labels. The LLM uses those same 750 labels to pick 8 examples and a threshold. Nothing else differs; only how the labels are used.

LLM ARM

750 labels → 8 prompt examples (4 OFF / 4 NOT) → 742 labels set the threshold → score dev as log P(A) − log P(B).

FINE-TUNING ARM

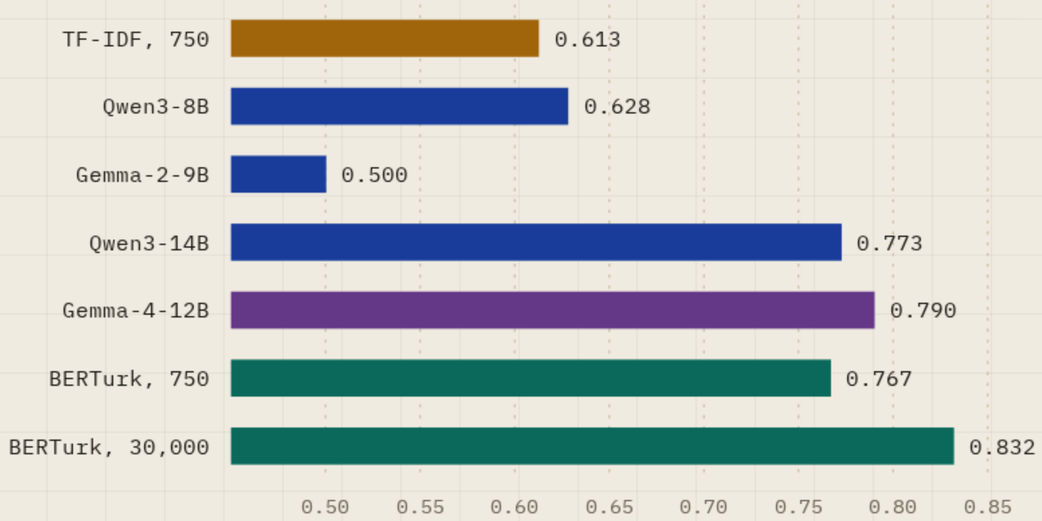
BERTurk and XLM-R-base, same subsets: 128 tokens, batch 16, lr 2e-5, 4 epochs or 400 updates, no early stopping. 46 runs.

| SYSTEM (DEV) | MACRO-F1 | RUNS | THRESHOLD |
|---|---|---|---|
| Qwen3-8B, zero-shot | 0.162 | 1 | 0 |
| Qwen3-8B, 8 + 742 | 0.628 ± 0.034 | 3 | 3.13 / 2.75 / 2.50 |
| Gemma-2-9B, zero-shot | 0.412 | 1 | 0 |
| Gemma-2-9B, 8 + 742 | 0.500 ± 0.048 | 3 | 0.63 / 0.25 / 0.88 |
| BERTurk, 750 | 0.767 ± 0.014 | 10 | n/a |
| BERTurk, 30,000 | 0.832 ± 0.003 | 3 | n/a |

READING THE DEV TABLE

Zero-shot to calibrated is +0.466 macro-F1 for Qwen3-8B and +0.088 for Gemma-2-9B. BERTurk-750 sits at 0.767. These are dev numbers; the test comparison is still pending.

# BERTurk beats both proposal LLMs; Gemma-4-12B leads on dev




| SYSTEM | DEV MACRO-F1 | RUNS |
|---|---|---|
| TF-IDF + LR, 750 | 0.613 ± 0.015 | 10 |
| Qwen3-8B | 0.628 ± 0.034 | 3 |
| Gemma-2-9B | 0.500 ± 0.048 | 3 |
| Qwen3-14B | 0.773 ± 0.009 | 3 |
| Gemma-4-12B | 0.790 ± 0.007 | 3 |
| BERTurk, 750 | 0.767 ± 0.014 | 10 |
| BERTurk, 30,000 | 0.832 ± 0.003 | 3 |

**+0.154**
BERTurk-750 vs TF-IDF

**+0.139**
vs Qwen3-8B

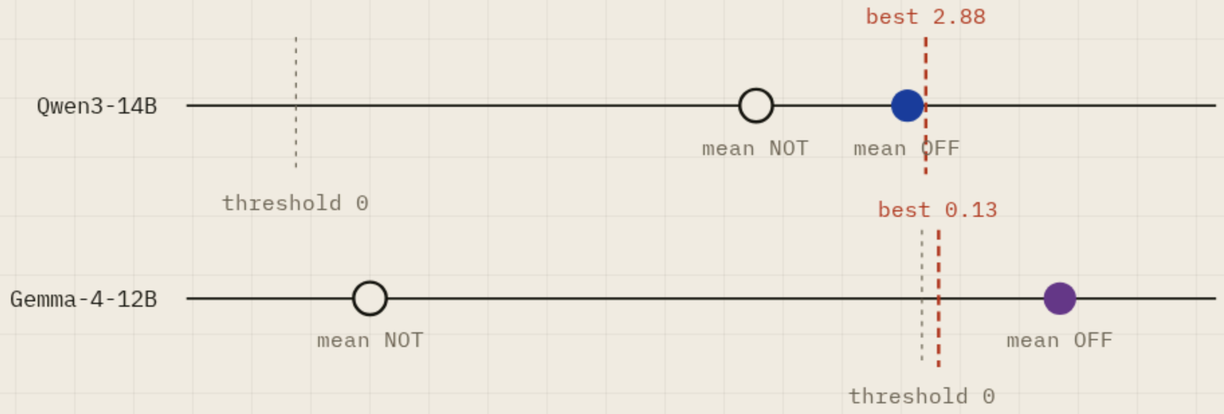
**+0.267**
vs Gemma-2-9B

**−0.023**
Gemma-4-12B ahead of BERTurk

Dev only. The frozen-test comparison is still pending.

# Qwen is shifted; Gemma's threshold is already near optimal

| ZERO-SHOT DIAGNOSTIC | F1 @ 0 | F1 @ BEST | AUC | MEAN OFF | MEAN NOT |
|---|---|---|---|---|---|
| Qwen3-14B | 0.171 | 0.677 | 0.756 | 2.79 | 2.10 |
| Gemma-4-12B | 0.763 | 0.765 | 0.864 | 1.03 | -4.13 |




Mean log P(A) − log P(B); zero and best thresholds marked. F1 @ 0 uses threshold 0.

WHAT EXPLAINS THE GAP

- **It is not the tokenizer.** Gemma packs these tweets more tightly: 32.3 vs 37.9 tokens/tweet.
- **Qwen is shifted, not blind.** At threshold 0 it calls 99% of tweets offensive, but the ranking still works: best-threshold F1 is 0.677.
- **Gemma is already centred.** F1 at 0 is 0.763 and the best threshold reaches 0.765, so threshold tuning has almost nothing to gain here.
- **The 8 examples still help Qwen:** 0.677 → 0.773 after calibration.

This is threshold geometry, not proof of contamination. It does not separate pretraining language from alignment; the exact-completion check is still planned.

# One dev pass on an L4: 3 to 5 tweets/s with examples, 10 to 18 without

| SYSTEM | PARAMS | DEV SCORING | THROUGHPUT | PEAK VRAM | NOTES |
|---|---|---|---|---|---|
| TF-IDF, 750 | n/a | 0.6 s | ~3,100/s | CPU | fit 1.0 s |
| TF-IDF, 30,000 | n/a | 0.6 s | ~3,100/s | CPU | fit 14.6 s |
| Qwen3-8B, few-shot | 8.2B | 355 s | 5.0/s | n/a | 3 runs |
| Gemma-2-9B, few-shot | 9.2B | 430 s | 4.1/s | n/a | 3 runs |
| Qwen3-14B, few-shot | 14.8B | 644 s | 2.7/s | n/a | 3 runs |
| Gemma-4-12B, few-shot | 12.0B | 579 s | 3.0/s | n/a | 3 runs |
| Qwen3-14B, zero-shot | 14.8B | 180 s | 9.8/s | 10.7 GB | load 157 s · 1,290 tok/s |

Scoring the 1,756 dev tweets; model download and load are excluded unless stated. Few-shot is 3 to 4× slower because each prompt carries 8 examples (~600 tokens).

BERTurk: 35 s training at 750 labels, ~29 min at 30,000. Gemma-4-26B-A4B exceeded 22 GB and could not load on L4. Reported runs: 9.4 Colab units (~2.5 L4-hours).

WHAT THIS BUYS

Few-shot costs 3 to 4× more per dev pass; BERTurk-750 trains in 35 s, so all 46 fine-tuning runs fit in about 3 Colab units. The LLM arm is the expensive part.

# Done: BERTurk and Ablation A. Every remaining step has a date and an owner

| DATE | EXPERIMENT | OWNER | STATUS |
|---|---|---|---|
| 11 Nov | BERTurk 750 (10) + 30,000 (3) on dev | Vesile · Mert | done |
| 11 Nov | Ablation A: BERTurk vs XLM-R, 750 | Vesile | done |
| 12 Nov | Mid-project talk; freeze this deck | All | talk |
| 21 to 24 Nov | Ablation B: Gemma with English instructions | Gürkan | planned |
| 25 to 27 Nov | Test-contamination check, first 200 test tweets | Gürkan · Utku | planned |
| 28 to 30 Nov | Freeze methods; open test; final runs | All | planned |
| 1 to 5 Dec | Paired bootstrap and error analysis | Mert · Utku | planned |
| 1 to 5 Dec | Paper draft | All | planned |
| 8 Dec | Final report | All | planned |

BIGGEST RISK

A 26B+ model may not fit the L4: Gemma-4-26B-A4B OOM'd at 22 GB, so we used the 12B dense model. **Fallback:** Qwen3-14B alone.

RESULT RISK

The lines may never cross. The 750-label question still has a clear answer; the learning curve is descriptive.

On proposal feedback: the rubric asks for test-set baseline numbers, so we scored the deterministic TF-IDF baseline once on test and kept methods on dev.

RULE

Methods freeze before the test set opens. The final answer uses the paired bootstrap; the learning curve stays descriptive.

# What we know, what we do not, and what comes next

BASELINE · TEST

## 0.763

TF-IDF test macro-F1 at 30,000 labels. At 750 labels it falls to 0.620 ± 0.012. Deterministic; test 95% CI [0.745, 0.779].

BEST LLM · DEV

## 0.790

Gemma-4-12B dev macro-F1 after calibration; zero-shot 0.763. Qwen3-14B reaches 0.773. Threshold from 742 labels; 3 runs.

FINE-TUNE · DEV

## 0.767

BERTurk-750 dev (10 runs); 30,000 labels reaches 0.832. It beats Qwen3-8B and Gemma-2-9B, but Gemma-4-12B leads on dev. Gap to Gemma-4-12B: −0.023.

**Still missing:** test-set scores for both arms. Everything above is dev; the final verdict uses the frozen 3,528-tweet test set. Four decisions were fixed before any method ran: the test split, macro-F1, the 0.02 minimum gap, and a paired bootstrap over 10,000 resamples to check that the gap is not chance.

Questions welcome; any member can answer any part.

# Title, abstract, research question, contributions

1 · TITLE AND ABSTRACT

**Fine-tuning BERTurk vs threshold-calibrated open LLMs for Turkish offensive language with 750 labels.**

Small Turkish platforms can label only a few hundred tweets. We ask a narrow question: with 750 labelled OffensEval-TR tweets, does a fine-tuned BERTurk beat an open LLM that gets the same 750 labels as 8 prompt examples and 742 calibration tweets for its decision threshold? The TF-IDF + logistic-regression baseline scores 0.763 macro-F1 on the frozen 3,528-tweet test set, but only 0.620 with 750 labels. The LLM arm scores each tweet as log P(A) − log P(B), and the 742 labels set its threshold. On dev, calibration moves Qwen3-8B from 0.162 to 0.628 and Gemma-2-9B from 0.412 to 0.500; Qwen3-14B reaches 0.773 (zero-shot 0.171) and Gemma-4-12B 0.790 ± 0.007 (zero-shot 0.763). Fine-tuned BERTurk reaches 0.767 ± 0.014 on dev with 750 labels. **Gap:** the frozen-test comparison is pending; the final result uses the paired bootstrap.

2 · RESEARCH QUESTION

Unchanged. With 750 labels, does fine-tuning BERTurk beat prompting Gemma-2-9B-it or Qwen3-8B (same 750 tweets, 8 examples plus 742 threshold labels) on macro-F1 over the official 3,528-tweet test set?

**"No" result:** BERTurk's mean over 10 runs is not ≥0.02 above the LLM's mean over 3 runs, or the paired bootstrap says chance ($p \geq 0.05$). **Change:** none.

3 · CONTRIBUTIONS

**C1.** At 750 labels, we test whether BERTurk's mean test macro-F1 is ≥0.02 above each LLM's, and whether a paired bootstrap rejects chance. Test pending; on dev BERTurk 0.767 vs 0.628 / 0.500.

**C2.** The 742-label threshold improves each LLM's dev macro-F1 over zero-shot; we report per-model delta and spread.

**C3.** The learning curve tests where fine-tuning overtakes TF-IDF between 250 and 6,000 labels; if the lines never cross, the 750-label answer stands.

# Related work and data

4 · RELATED WORK

1. **Zampieri et al. 2020.** “SemEval-2020 Task 12: Multilingual Offensive Language Identification in Social Media (OffensEval 2020).” SemEval. Top Turkish systems fine-tune on full data; no label-budget comparison.
2. **Çöltekin 2020.** “A Corpus of Turkish Offensive Language.” LREC. The dataset; no fine-tuning-vs-prompting comparison.
3. **Zampieri et al. 2023.** “OffensEval 2023: Offensive Language Identification in the Age of Large Language Models.” NLE. Zero-shot LLMs; Turkish only Flan-T5-large 0.451; no examples.
4. **Dehghan & Yanıkoğlu 2024.** “Evaluating ChatGPT’s Ability to Detect Hate Speech in Turkish Tweets.” CASE, pp. 54-59. ChatGPT with 7 examples vs BERTurk, one dataset and one size.
5. **Pecher, Srba & Bielikova 2025.** “Comparing Specialised Small and General Large Language Models on Text Classification.” EMNLP, pp. 165-184. English break-even near 100 labels; we test noisy Turkish two-class data.
6. **Schmitt & Schweter 2025.** “SindBERT.” arXiv:2510.21364. Turkish encoders, full data only; no LLMs or small budgets.
7. **Şeker, Uludoğan, Önal & Özgür 2025.** “HATECAT-TR.” Findings of EMNLP. Turkish span hate; BERTurk 0.59 vs LLaMA-3.3-70B 0.53; we classify whole tweets.
8. **Karakaş & Şimşek 2026.** arXiv:2606.29614. Zero-shot LLMs vs BERTurk at one size; we add examples, calibration and a curve.

5 · DATA

OffensEval-TR 2020, Çöltekin, LREC 2020; annotations CC BY 2.0; Turkish; @USER/URL masked. No extra cleaning.

| SPLIT | TWEETS | OFF | OFF % |
|---|---|---|---|
| Official train | 31,756 | 6,131 | 19.3 |
| Our train | 30,000 | 5,792 | 19.3 |
| Dev | 1,756 | 339 | 19.3 |
| Test (frozen) | 3,528 | 716 | 20.3 |

9 duplicate train texts; 1 test/train overlap. Example (OFF, id 33518): “Kırmızı ağır karar Allah belanı versin”.

# Method, protocol, results so far, error analysis

### 6 · METHOD AND BASELINE

TF-IDF word 1-2 + char_wb 2-5, min_df=2, sublinear, Turkish lowercasing; LR C=1.0, balanced. Test 0.763 [0.745, 0.779].

Fine-tuning: BERTurk/XLM-R, 128 tokens, batch 16, 2e-5, 4 epochs or 400 updates, class-weighted, no early stopping. LLM: Qwen/Gemma 4-bit, 8 examples, 742-label threshold. One change: labels train weights vs choose examples and threshold.

Compute used: about 3 Colab units for the 46 fine-tuning runs + 2.5 L4-h for 16 LLM passes; about 18 GPU-h needed for the final.

### 7 · EVALUATION PROTOCOL

Macro-F1; frozen 3,528-tweet test; 10 BERTurk runs at 250/750/2,000 and 3 at 6,000/30,000; 3 runs per LLM at 750. Paired bootstrap, 10,000 resamples; real gap ≥0.02 and wins on ≥95%. Fixed before method results; dev for code checks only.

### 8 · RESULTS SO FAR (MEAN ± SD)

| SYSTEM | DEV | TEST |
|---|---|---|
| Always NOT | 0.447 | 0.444 |
| TF-IDF, 750 | 0.613 ± 0.015 | 0.620 ± 0.012 |
| TF-IDF, 30,000 | 0.763 | 0.763 |
| Qwen3-8B zero | 0.162 | n/a |
| Qwen3-8B cal | 0.628 ± 0.034 | n/a |
| Qwen3-14B zero | 0.171 | n/a |
| Qwen3-14B cal | 0.773 ± 0.009 | n/a |

| SYSTEM | DEV | TEST |
|---|---|---|
| Gemma-2-9B zero | 0.412 | n/a |
| Gemma-2-9B cal | 0.500 ± 0.048 | n/a |
| Gemma-4-12B zero | 0.763 | n/a |
| Gemma-4-12B cal | 0.790 ± 0.007 | n/a |
| BERTurk, 250 | 0.680 ± 0.052 | n/a |
| BERTurk, 750 | 0.767 ± 0.014 | n/a |
| BERTurk, 2,000 | 0.793 ± 0.006 | n/a |
| BERTurk, 6,000 | 0.806 ± 0.004 | n/a |
| BERTurk, 30,000 | 0.832 ± 0.003 | n/a |
| XLM-R, 750 | 0.670 ± 0.025 | n/a |

### 9 · ERROR ANALYSIS: 60 AUDITED OF 275

False alarms: no insult 9; political/religious 10; sports 5; personal 3; sarcasm 1; label noise 2. Misses: implicit insult 8; masked profanity 5; veiled threat 3; political/religious 4; institutional/sports 3; label noise 7. One example per category is in results/errors/audit.json.

# Research lead, plan and risk, reproducibility

10 · RESEARCH LEAD SECTION

**Vesile Han (CSE 654), fine-tuning.** Pecher et al., EMNLP 2025: English break-even near 100 labels, more for two-class tasks and variance; we test Turkish noisy two-class. Schmitt & Schweter, SindBERT 2025: Turkish encoders, full data only; no LLMs. Ablation A (done): BERTurk 0.767 vs XLM-R 0.670, +0.097 [0.080, 0.116], 100% wins. KUISAIL published BERTurk 0.816 on test; our 30k run should reproduce.

**Gürkan Göztepeli (CSE 654), prompting.** Şeker et al., HATECAT-TR 2025: BERTurk 0.59 vs LLaMA-3.3-70B 0.53 on Turkish hate spans; we classify whole tweets and score probabilities. Karakaş & Şimşek 2026: zero-shot LLMs vs BERTurk at one size; we add examples, calibration, curve. Ablation B: Gemma English instructions, 3 runs.

11 · PLAN AND RISK

Done: BERTurk curve (36 runs), Ablation A (10). 21 to 24 Nov: Ablation B (Gürkan). 25 to 27 Nov: contamination check (Gürkan). 28 to 30 Nov: freeze, open test, final runs (all). 1 to 5 Dec: bootstrap and error analysis (Mert · Utku), paper draft (all). 8 Dec: report (all).

**Biggest risk:** 26B+ may not fit L4; Gemma-4-26B OOM'd at 22 GB, so 12B used. Fallback: Qwen3-14B alone. **Result risk:** lines may never cross; 750-label answer stands. **Proposal feedback:** add a test-set baseline; it is reported once, while methods stay on dev.

12 · REPRODUCIBILITY

Repository: github.com/paysis/offenseval-tr-midproject. One command: `bash scripts/reproduce.sh gpu`. Seeds: split 42, subsets 42+k, bootstrap 42. Versions: Python 3.13.16, scikit-learn 1.9.1, transformers 5.22.1, torch 2.14.1, bitsandbytes 4-bit NF4.

**Roles:** fine-tuning side, Vesile Han and Mert Uğur; prompting side, Gürkan Göztepeli and Utku Gökçek. Ablation A belongs to Vesile, Ablation B to Gürkan. **AI tools:** a coding assistant was used for code, analysis and drafting; every number is read from saved result files and checked by the team.